It is straightforward to show that each of the terms of the corresponding vector field splitting preserves the first integral $\Lambda =$ constant. For example

$$\begin{aligned}\mathcal{L}_{\mathrm{F}}\Lambda &= \left[F^{\mathrm{F}} - (p^2/\Lambda)F^{\mathrm{F}}\right]\frac{\partial \Lambda}{\partial p} - \frac{p\xi_1 F^{\mathrm{F}}}{\Lambda}\frac{\partial \Lambda}{\partial \xi_1} \\ &= \left[F^{\mathrm{F}} - (p^2/\Lambda)F^{\mathrm{F}}\right](2p) - \frac{p\xi_1 F^{\mathrm{F}}}{\Lambda}(\mu_1\xi_1) \\ &= pF^{\mathrm{F}}\left[2 - \Lambda^{-1}(2p^2 + \mu_1\xi_1^2)\right] \\ &= 2pF^{\mathrm{F}}\left[1 - \Lambda^{-1}\Lambda\right] = 0.\end{aligned}$$

The proof for the other terms is similar.

The given decomposition can be used to construct a variety of algorithms, for example the following which we term the Stochastic Isokinetic Nosé-Hoover RESPA algorithm.

SIN(R) Algorithm

Propagate p, ξ_1, ξ_2 by $\exp\left[(h/2)\mathcal{L}_{\mathrm{N}}\right]$.
Propagate p, ξ_1 by $\exp\left[(\tilde{h}/2)\mathcal{L}_{\mathrm{F}} + (h/2)\mathcal{L}_{\mathrm{S}}\right]$, $\quad (\tilde{h} = h/r)$.
Propagate q by $\exp\left[\tilde{h}\mathcal{L}_{\mathrm{K}}\right]$.
Propagate ξ_2 by $\exp\left[\tilde{h}\mathcal{L}_{\mathrm{O}}\right]$.
Propagate p, ξ_1 by $\exp\left[(\tilde{h}/2)\mathcal{L}_{\mathrm{F}}\right]$.
Repeat $r - 2$ times:
  Propagate p, ξ_1 by $\exp\left[(\tilde{h}/2)\mathcal{L}_{\mathrm{F}}\right]$.
  Propagate q by $\exp\left[(\tilde{h}/2)\mathcal{L}_{\mathrm{K}}\right]$.
  Propagate ξ_2 by $\exp\left[(\tilde{h})\mathcal{L}_{\mathrm{O}}\right]$.
  Propagate q by $\exp\left[(\tilde{h}/2)\mathcal{L}_{\mathrm{K}}\right]$.
  Propagate p, ξ_1 by $\exp\left[(\tilde{h}/2)\mathcal{L}_{\mathrm{F}}\right]$.
Propagate p, ξ_1 by $\exp\left[(\tilde{h}/2)\mathcal{L}_{\mathrm{F}}\right]$.
Propagate ξ_2 by $\exp\left[\tilde{h}\mathcal{L}_{\mathrm{O}}\right]$.
Propagate q by $\exp\left[\tilde{h}\mathcal{L}_{\mathrm{K}}\right]$.
Propagate p, ξ_1 by $\exp\left[(h/2)\mathcal{L}_{\mathrm{F}} + (h/2)\mathcal{L}_{\mathrm{S}}\right]$.
Propagate p, ξ_1, ξ_2 by $\exp\left[(h/2)\mathcal{L}_{\mathrm{N}}\right]$.